BIOSTATISTICS

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Biostat scares everyone
but Pihu makes it EASY.
Trust me on this one!

Biostatistics | PSM | 3rd Year | MBBS

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Step 1: Know your DATA type. Everything else follows from this!

DEFINITION

"The science of collecting, summarizing, analyzing & interpreting data in medicine & public health."

Types of Data (Stevens' Classification)

Scale	Type	Order?	Central Tend.	Examples
Nominal	Qualitative	No	Mode	Blood group, Gender, Religion
Ordinal	Qualitative	Yes	Median	Pain severity, SES, Likert scale
Interval	Quantitative	Yes	Mean	Temperature (C/F), IQ score
Ratio	Quantitative	Yes	Mean	Height, Weight, BP, Hb level

★ Mean & SD can ONLY be calculated for Interval & Ratio data. Mode only for Nominal!

Measures of Central Tendency

Measure	Formula / How	Merits	Demerits
Mean	Sum of xi / n	Uses all values, stable	Affected by outliers
Median	Middle value (n+1)/2	Not affected by outliers	Ignores most values
Mode	Most frequent value	Works for nominal data	May not exist / unstable

★ Key Formula — asked EVERY year! ★

Mode = 3(Median) - 2(Mean)

Positive skew: Mean > Median > Mode | Negative skew: Mean < Median < Mode

Symmetrical: Mean = Median = Mode

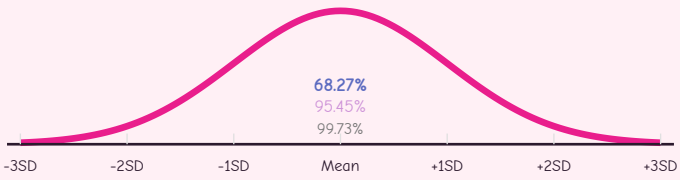
The bell curve is THE most drawn diagram in PSM exams. DRAW IT!

Measures of Dispersion

Measure	Formula	Key Point
Range	Max - Min	Simplest, uses only 2 values
Variance	Sum(xi-mean)^2 / (n-1)	Sample uses n-1 (Bessel's correction)
SD	Square root of Variance	BEST measure, uses all values
SE	SD / sqrt(n)	Precision of sample mean estimate
CV	(SD / Mean) x 100	Compares variability across datasets

★ **SD vs SE — Classic SAQ:**
SD = variability within ONE sample (independent of n)
SE = precision of sample mean (decreases as n increases, SE = SD/sqrt n)

Normal (Gaussian) Distribution



Properties:

- Bell-shaped, symmetrical
- Mean = Median = Mode
- Area under curve = 1
- Asymptotic (never touches x-axis)
- Defined by Mean & SD only

★ **THE 68-95-99.7 Rule — MUST MEMORISE!**
Mean +/- 1SD = 68.27% | Mean +/- 2SD = 95.45% | Mean +/- 3SD = 99.73%
Z-score = (X - Mean) / SD (converts any normal dist to standard normal)

Sampling Methods

Method	How It Works	When to Use
Simple Random	Lottery / random number table	Small homogeneous population
Systematic	Every kth individual (k=N/n)	Population in a list
Stratified	Divide into strata, sample each	Heterogeneous population
Cluster	Random entire clusters, study all	Large areas, no listing
Multistage	Random clusters then random people	National surveys (NFHS)

Which test to use?
THIS table is asked
as LAQ + SAQ both!

Which Statistical Test? (The Money Table)

Scenario	Data	Groups	Test
1 mean vs known value (small n)	Quant.	1	One-sample t-test
2 indep. group means (n<30)	Quant.	2 indep.	Unpaired t-test
2 indep. group means (n>30)	Quant.	2 indep.	Z-test
Before/after SAME group	Quant.	2 paired	Paired t-test
3+ group means	Quant.	3+	ANOVA (F-test)
2 categorical variables	Qualit.	2 vars	Chi-square test
2 proportions (large n)	Qualit.	2	Z-test for proportions
Non-normal, 2 independent	Ordinal	2	Mann-Whitney U
Non-normal, 2 paired	Ordinal	2	Wilcoxon signed-rank
Non-normal, 3+ groups	Ordinal	3+	Kruskal-Wallis

Chi-Square Test (Very High Yield LAQ)

Formula:

X^2 = Sum of (O - E)^2 / E

df = (rows-1) x (cols-1)

E = (Row total x Col total) / Grand total

If E < 5: use Yates' correction or Fisher's

t-test Key Facts:

Used when n < 30, pop. SD unknown

Unpaired df = n1+n2-2

Paired df = n-1

"Before-after on same pts" = Paired

p-value, CI + Errors

Concept	What You Need to Know
p-value < 0.05	Statistically significant — reject null hypothesis (H0)
95% CI	Mean +/- 1.96 x SE | If CI excludes null value = significant
99% CI	Mean +/- 2.58 x SE
Type I (alpha)	Reject H0 when true = FALSE POSITIVE (set at 0.05)
Type II (beta)	Accept H0 when false = FALSE NEGATIVE (set at 0.20)
Power	1 - beta = 0.80 = probability of detecting true effect

Correlation does NOT imply causation!
Write this in every Q!

Correlation

Type	When to Use	Key Point
Pearson's r	Normal continuous data	Parametric, r ranges -1 to +1
Spearman's rho	Ordinal / non-normal data	Non-parametric, rank-based

r = +1: Perfect positive | r = -1: Perfect negative | r = 0: No correlation
r^2 = Coefficient of determination = proportion of variance explained

Regression

Linear Regression: Y = a + bX
a = intercept (Y when X=0) | b = slope (regression coefficient)
Predicts Y from X | NOT scale-free (unlike correlation)

Sensitivity + Specificity (The 2x2 Table)

	Disease +	Disease -
Test +	a (TP)	b (FP)
Test -	c (FN)	d (TN)

Sensitivity = a/(a+c)
Specificity = d/(b+d)
PPV = a/(a+b)
NPV = d/(c+d)

★ SnNOut / SpPIn — NEVER forget these! ★
SnNOut: High Sensitivity + Negative result = rules OUT disease
SpPIn: High Specificity + Positive result = rules IN disease
Sn & Sp are TEST properties (don't change with prevalence). PPV & NPV DO change!

Prevalence UP = PPV UP, NPV DOWN | Prevalence DOWN = PPV DOWN, NPV UP

Exam Strategy + High Yield

You've survived biostats!
These Qs are 100%
predictable. MEMORISE!

Most Repeated LAQs (10-12 marks)

1. Normal distribution — define, properties, diagram, 68-95-99.7 rule, applications
2. Measures of central tendency — Mean, Median, Mode, formulas, merits/demerits
3. Tests of significance — types (chi-sq, t-test, Z, ANOVA), when to use each
4. Chi-square test — formula, steps, df, Yates' correction, worked example
5. Sampling methods — probability vs non-probability, types, merits/demerits

Most Repeated SAQs (5-6 marks)

1. SD vs SE — definition, formula, key differences (table format!)
2. Types of data — Nominal, Ordinal, Interval, Ratio + examples
3. p-value and statistical significance
4. 95% Confidence Interval — formula and interpretation
5. Type I vs Type II errors
6. Paired vs Unpaired t-test — when to use
7. Sensitivity & Specificity from 2x2 table

Must-Know Formulas

Formula	Expression
Mean	Sum(xi) / n
Mode (empirical)	3(Median) - 2(Mean)
Variance (sample)	Sum(xi-mean)^2 / (n-1)
SD	sqrt(Variance)
SE	SD / sqrt(n)
CV	(SD/Mean) x 100
95% CI	Mean +/- 1.96 x SE
Chi-square	Sum of (O-E)^2 / E
Z-score	(X - Mean) / SD

High Yield Checklist

MUST KNOW

- Normal dist. + bell curve
- Mean/Median/Mode formulas
- SD vs SE difference
- Chi-square formula + df
- Sensitivity/Specificity

HIGH YIELD

- Which test to use table
- p-value interpretation
- 95% CI formula
- Type I & II errors
- Paired vs unpaired t-test

BONUS

- Kruskal-Wallis test
- Spearman's correlation
- ANOVA post-hoc tests
- Fisher's exact test

♥ **Biostatistics is just maths with a stethoscope, Sammy!**
You already know more than you think. Go ace this paper!